

Important Questions

Q1.

- (a) What is a time series? Differentiate between univariate and multivariate time series with examples. Why is forecasting essential for business and economic planning?
- (b) The following table shows monthly sales (in Rs. thousands):
Jan–Jun = [120, 135, 150, 160, 145, 155].
Compute the 3-month moving average and briefly interpret how smoothing affects the visibility of trend and random fluctuations.

Q2.

- (a) Explain the components and general structure of the ARIMA(p, d, q) model. What roles do parameters p, d , and q play in capturing time series patterns?
- (b) For the following non-stationary series, perform the necessary differencing to achieve stationarity and identify a possible ARIMA model form:

$$Y_t = Y_{t-1} + 0.7Y_{t-2} - 0.2Y_{t-3} + \varepsilon_t$$

- (c) Discuss the use of Akaike Information Criterion (AIC) in model selection. How does it balance model complexity and goodness of fit?

Q3.

- (a) Differentiate between the Yule-Walker estimation and Maximum Likelihood Estimation (MLE) for ARMA parameters. Why is MLE generally preferred for models containing an MA component?
- (b) A time series is modeled as an MA(2) process: $Y_t = \epsilon_t + 0.6\epsilon_{t-1} - 0.3\epsilon_{t-2}$. If $\epsilon_4 = 2$ and $\epsilon_5 = 1.5$, and all prior error terms are zero, calculate the value of Y_5 .

Q4.

- (a) Explain how a Vector Autoregressive (VAR) model captures inter-series dependencies. State the stability condition for a VAR(1) model.
- (b) Given the VAR(1) system:
$$\begin{bmatrix} X_t \\ Y_t \end{bmatrix} = \begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.7 \end{bmatrix} \begin{bmatrix} X_{t-1} \\ Y_{t-1} \end{bmatrix} + \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{bmatrix}$$
 where $X_{t-1} = 120$, $Y_{t-1} = 100$, $\epsilon_{1t} = 2$, $\epsilon_{2t} = -1$, compute X_t and Y_t .
- (c) Explain the main differences between a VAR(p) model and a VARMA(p, q) model. Why is including a moving average (MA) component useful in some cases?

Q5.

- (a) State the mathematical equations for the additive and multiplicative decomposition models. What is the key assumption about the seasonal component in each model?
- (b) Given the quarterly data: $[30, 40, 50, 35, 45, 55]$. Calculate the 4-quarter centered moving average value corresponding to the 3rd quarter.

Q6.

- (a) Define weak stationarity. Explain how differencing can transform a non-stationary series into a stationary one with a suitable example.
- (b) For an AR(1) process defined as

$$Y_t = 0.8Y_{t-1} + \varepsilon_t,$$

where $\varepsilon_t \sim N(0, 1)$, compute the mean, variance, and autocorrelation function (ACF) at lag 1 and lag 2.

- (c) Discuss the Yule–Walker equations and their role in parameter estimation of AR processes.

Q7.

- (a) Define a stationary process. Describe how differencing and the Augmented Dickey–Fuller (ADF) test are used to detect and remove non-stationarity.

- (b) For the process $Y_t = 0.5Y_{t-1} + \epsilon_t + 0.4\epsilon_{t-1}$ where ϵ_t is white noise with variance 1, determine whether the process is stationary and compute its first two autocorrelations.

Q8.

- (a) Define Time Series. Explain the difference between univariate and multivariate time series with suitable real-world examples.
- (b) A time series Y_t is given by the following quarterly data (in units):

120, 135, 150, 165, 130, 145, 160, 175

Decompose the series into Trend and Seasonal components using the Additive model. Clearly show all calculation steps.

- (c) Discuss the significance of Trend, Seasonality, and Irregular components in forecasting accuracy. How does decomposition aid in model selection?

Q9.

- (a) Identify one major limitation of traditional VAR/VARMA models in capturing temporal dependencies. Explain how a deep learning model such as an LSTM addresses this limitation through its architecture and memory mechanism.
- (b) For a multivariate time series with 5 features and a look-back window of 10 time steps, describe and contrast the input data representation required for forecasting using a VAR model and an LSTM network.

Q10.

- (a) Differentiate between ARIMA and Seasonal ARIMA (SARIMA) models. Write the general form of a $SARIMA(p, d, q)(P, D, Q)_s$ model and briefly explain the significance of each parameter.
- (b) A quarterly sales series exhibits strong seasonality. Assuming a seasonal period of four, propose an appropriate SARIMA model structure and outline the complete process of model building — identification, estimation, diagnostic checking, and forecasting.
- (c) Discuss how residual characteristics such as autocorrelation, normality, and homoscedasticity affect forecast reliability. Specify the diagnostic tests used to verify the adequacy of a fitted SARIMA model.

Q11.

- (a) Differentiate between non-seasonal ARIMA and Seasonal ARIMA (SARIMA) models. Write the general SARIMA(p, d, q)(P, D, Q)_s formulation and explain each parameter.
- (b) Two competing SARIMA models for quarterly data yield the following:

Model	AIC	RMSE
SARIMA(1,1,1)(1,1,0) ₄	215.3	12.4
SARIMA(0,1,2)(0,1,1) ₄	210.8	13.1

Which model would you select and why? Discuss how information criteria complement residual diagnostics.

Q12.

- (a) What is weak-sense stationarity? Explain how the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) are used together to identify the order of an ARMA model.
- (b) An AR(1) process is given by $Y_t = 0.8Y_{t-1} + \epsilon_t$. Calculate the theoretical autocorrelation values for lag 1, lag 2, and lag 3 (i.e., find ρ_1, ρ_2, ρ_3).

Q13.

- (a) Explain the difference between ACF and PACF plots. How are they used in identifying the order of AR and MA components in ARMA(p, q) models?
- (b) Given the following ACF and PACF patterns, identify the most appropriate model type (AR, MA, or ARMA) and justify your answer:

Lag	1	2	3	4	5
ACF	0.75	0.50	0.25	0.10	0.05
PACF	0.75	0.05	0.02	0.01	0.00

- (c) Briefly explain the Maximum Likelihood Estimation (MLE) approach for ARMA model parameter estimation and discuss one key advantage over Yule–Walker estimation.

Q14.

- (a) State the primary condition under which a Vector Autoregression (VAR) model is more appropriate than fitting several independent ARIMA models. What is the purpose of the Granger Causality test in a VAR context?
- (b) Write the complete matrix equation for a **VAR(1)** model with two endogenous variables, $y_{1,t}$ and $y_{2,t}$. Clearly define all vectors and matrices used.

Q15.

- (a) Compare the working principles of ARIMA, VAR, and LSTM models in capturing temporal dependencies. Discuss their relative strengths and weaknesses.
- (b) Why might a Transformer-based deep learning architecture be preferred over a traditional RNN or LSTM for a very long-range forecasting task?.

Q16.

- (a) Derive the general equation for a k-point Moving Average method. Illustrate with an example using a 3-point moving average for the series:

42, 44, 46, 48, 50, 52

- (b) Explain the concept of Simple Exponential Smoothing and compute the forecast for period $t = 6$ given:

$$Y_t = [50, 52, 54, 53, 55], \quad \alpha = 0.3$$

Use the initial forecast $F_1 = 50$. Show all steps.

- (c) Compare Holt's Linear Trend Model and Holt-Winters Method in terms of applicability, parameter estimation, and handling of seasonality.

Q17.

- (a) In what specific scenarios would you choose Holt's model over Simple Exponential Smoothing, and Holt-Winters' model over Holt's model?
- (b) For a time series, the level at period $t - 1$ is $L_{t-1} = 150$ and the trend is $T_{t-1} = 5$. Given the observed value $Y_t = 158$, calculate the updated level L_t and trend T_t using Holt's model equations with $\alpha = 0.5$ and $\beta = 0.4$.

Q18.

- (a) Using Simple Exponential Smoothing with $\alpha = 0.4$ and initial forecast $F_1 = 150$, forecast week 6 for the following weekly sales data: [150, 155, 162, 158, 165]. Show all recursive calculations.
- (b) Explain the limitations of Simple Exponential Smoothing and describe how Holt's Double Exponential Smoothing overcomes them. Derive both smoothing equations for level and trend.

Q19.

- (a) Describe the structure of a Vector Autoregressive (VAR) model and explain how it differs from univariate AR models.
- (b) Given two time series X_t and Y_t , formulate a VAR(1) model and explain the interpretation of coefficients in terms of causality and interdependence.
- (c) Discuss how Deep Learning models such as LSTM and Seq2Seq networks can enhance forecasting accuracy in multivariate time series. Highlight their advantages and limitations compared to traditional VARMA models.

Q20.

- (a) Outline the step-by-step approach to identifying and estimating an ARIMA(p,d,q) model. Explain the significance of ACF/PACF plots in deciding p and q .
- (b) Discuss the Yule–Walker equations and their role in parameter estimation of AR processes.

Q21.

- (a) Explain the roles of the differencing parameters d (non-seasonal) and D (seasonal) in a SARIMA(p, d, q)(P, D, Q) $_m$ model.
- (b) An analyst fits an ARIMA(1,1,0) model to data, obtaining the equation $(1 - 0.5B)(1 - B)Y_t = \epsilon_t$. If $Y_{t-1} = 100$ and $Y_{t-2} = 95$, what is the one-step-ahead forecast for Y_t ?